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Project Proposal
CSc 84030 Big Data Analytics
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Project Proposal – Spark streaming pipeline for Twitter data analytics.

Key Idea: Using real-time streaming data from Twitter to graph important figures (people's) relationships.

Since covid, the world as a whole turned to social media where platforms are constantly output streaming data every second. The data streams from these platforms don't stop and are very useful to tell what is happening in the world around us. Frameworks like Apache are now a core part of these data ecosystems with reading, processing and storing a lot of data really fast.

In this project, I propose to use distributed streaming frameworks to analyze Twitter conversations mentioning trending people as it occurs live. The goal is to mix sentiment detection with aggregation to gage both how users feel about these important people plus how they relate to each other. The goal is to show how live big data analytics can extract patterns from live social streams using fault tolerant and scalable infrastructure.

The work planned is broken down as the following:

- Ingest data using Twitter's API while only looking for key profiles or key words. Each tweet will be pushed into a Kafka topic using python.
- Process data stream with Apache Spark park by consuming raw tweets from Kafka, parse them into a clean data format, identify important/key people by name matching, use a simple sentiment model to classify the data, then make a graph with relationship edges.
- The data will be stored in Apache Parquet for durability and offline queries.
- The system will be deployed with a Kafka broker, and Spark driver plus workers with Docker containers on a cloud VM.
- The project can be measured through how much data is going through the pipeline (latency) and the inferred data using sentiment analysis.
- If time permits, an open source dashboard (python friendly) can be made to have a visual of the data.

This project is to use a streaming pipeline with Apache Kafka and Apache Spark that will be able to process Twitter live data continuously. I expect this to produce sentiment trends for important figures (of my choosing but can be changed) with live updates as the data comes in. Performance of the system can be checked with streaming metrics like latency, throughput and fault recovery. This work is to show how big data analytics can be applied to socially relevant events and people.